

Publication Bias in Statistical Meta-Analysis

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May 1, 2023



Table of Contents

Executive Summary.....	3
Introduction.....	4
Statistical Meta-Analysis.....	6
Preliminary Data Exploration.....	6
Fixed-Effect Meta-Analysis.....	7
Random-Effects Meta-Analysis.....	8
Meta-Regression.....	11
Publication Bias.....	13
Funnel Plots.....	13
Egger's Regression Test.....	14
Trim-and-Fill Method.....	16
File-Drawer Method.....	17
Discussions and Conclusion.....	18
References.....	19
Appendix.....	20

Executive Summary

The primary objective of this report is to explain, demonstrate, and analyze the effects of publication bias in statistical meta-analysis. This was accomplished by examining a dataset regarding the effectiveness of cognitive behavioral therapy (CBT) on reducing recidivism in juvenile and adult offenders. The effect size at hand is the log-transformed odds ratio between non-treated recidivists and treated recidivists. Prior to assessing publication bias, both fixed-effect and random effects meta-analysis models were implemented. Both models returned a significant effect size greater than zero, suggesting that CBT is effective in reducing recidivism as the proportion of non-recidivists was higher in the treatment group than in the control group. Significant heterogeneity, or between-study variability, was detected so the fixed effect model was determined to be more appropriate for the dataset. In addition to this, meta-regression was performed in an attempt to explain this heterogeneity. This found that many covariates included in the dataset were insignificant in affecting the odds ratio aside from the anger control component, which was shown to significantly reduce the odds of recidivism. The insignificance of many of the covariates was likely due to multicollinearity between them. Afterward, publication bias was explored. This revealed a considerable amount of asymmetry in the funnel plot, and a significant intercept parameter for Egger's Regression test. Adjustments were therefore made to the dataset to address the existence of publication bias and to make the results of the random effects model more reliable. These adjustments were executed using the trim-and-fill method. Ultimately, the adjusted dataset was fitted to another random effects model, still resulting in a significant overall effect. Interestingly, the file-drawer method revealed that a very large number of insignificant studies would have to be included to change the overall result to insignificant. In other words, there is overwhelming evidence that CBT reduces recidivism. Thus, the final conclusion of the problem in context is that CBT is very effective in reducing the odds of recidivism, even with publication bias accounted for and in spite of several studies with results to the contrary. Though the dataset in question is not majorly affected by publication bias, the report still serves as a demonstration of the methods used to identify and address it.

I - Introduction

An important topic of discussion, especially in recent years, is the effectiveness of criminal rehabilitation. Namely, whether cognitive behavioral therapy (CBT) can reduce the recidivism in juvenile and adult offenders. Recidivism is defined as the tendency of a convicted criminal to reoffend. Thus, the purpose of CBT is to reduce the chances of a criminal reoffending the law, indicating effective rehabilitation. A number of studies have been conducted regarding the validity of this hypothesis, but with inconsistent results. While some studies conclude a significant difference in recidivism probability between CBT-treated offenders and non-treated offenders, others have not. This brings about the need for statistical meta-analysis, which can be used in an attempt to reconcile these inconsistencies by developing an overall estimate.

Meta-analysis, in its simplest terms, is an answer to the problem caused by inconsistent results between several studies examining the same research question. For example, if some studies find rainfall to be highly correlated with commute time, but others find no correlation, which is the *correct* conclusion? Statistical meta-analysis attempts to make sense of these conflicting results in order to reach an overall conclusion. The analysis uses the studies themselves as its sample instead of individuals or specimens as seen in conventional studies. Meta-analysis is essentially a study of studies, or an analysis of analyses, in that sense. The unifying factor between the studies, and the primary variable of interest in meta-analysis is the effect size. Each study utilized in a meta-analysis must have the same calculation of effect size so that they can be compared. There are various possible effect sizes which measure the relationship or difference between groups for each study, though only one will be discussed in this report as it relates to the topic.

The selected dataset, entitled *Landenberger (2005)* after its curator and year of publication, contains results from 58 studies on the effectiveness of CBT on reducing recidivism. To assess said effectiveness, each study provided the number of individuals in the four groups: re-offenders in the control (untreated) group (A), non-re-offenders in the control group (B), re-offenders in the CBT group (C), and non-re-offenders in the CBT group (D). These values will be used to calculate the odds ratio between the untreated and treated groups. Then, in order to improve the interpretability of the results, the ratio will be log-transformed with the natural log. This final value will serve as the effect size for the analysis, with values greater than zero suggesting a lower proportion of re-offenders in the CBT group than in the control group. Hence, values less than zero suggest a higher proportion of re-offenders in the CBT group. The formula for odds ratio, along with its log-transformation and standard error, are shown below.

$$OR = \frac{A*D}{B*C}, \log(OR) = \log\left(\frac{A*D}{B*C}\right), SE(\log(OR)) = \sqrt{\frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D}}$$

The variance of the effect size is absolutely crucial in meta-analysis, as it is paramount to the calculation of the overall effect size. That is, studies with lower variances will have a greater weight on the pooled estimate. Precision of the studies also ties in closely with publication bias, the primary focus of this report, which will be discussed in thorough detail.

The dataset was selected for several reasons. The first, and perhaps most important, is the large number of studies it involves. A large sample size is crucial to virtually any statistical analysis in that the assumptions of the central limit theorem can be met. That is, with a sample of 58 studies, it can be reasonably assumed that their effect sizes follow a normal distribution. Additionally, the dataset is thorough and detailed, with several potential covariates included such as length of session, demographics, and binary variables indicating the use of specific CBT

components. These variables could be used as moderators in meta-regression, a topic to be discussed in a later section of the report. Further increasing the utility of the data is that it is mostly complete with few missing values, meaning the sample size does not need to be largely reduced when exploring any of the variables. Lastly, the data relates to a pertinent societal subject matter. It will certainly be useful to gain a thorough understanding of the effectiveness of CBT on criminal rehabilitation, especially because these results may provide insight toward either topic in a broader sense.

With the background information established, the primary focus of the study can be expounded upon. One of the primary difficulties pervading statistical meta-analysis is the availability of data. It is extremely common for meta-analyses to lack a sufficient number of studies to conduct meaningful analysis. While this could be due to a number of reasons, including the possibility of a niche area of study with little research being conducted, the primary concern is publication bias. The key assumption before conducting meta-analysis (and statistical analyses in general) is that the collected data is comprehensive or at least representative. Publication bias arises when there is any belief that this assumption is not true. Most of the time, publication bias takes the form of selection bias, which is the tendency of researchers to only publish statistically significant or desirable results. The issues resulting from this are almost trivial, as the choice to not publish *insignificant* results directly leads to a disproportionate amount of *significant* results, skewing meta-analysis results and invalidating conclusions. While it is certainly possible for every conducted study to be significant, it is often unreasonable to assume this as nearly every sample of sufficient size can be expected to have at least one outlying result. Even if this were the case, statistical meta-analysis would be unnecessary as there is already significant evidence supporting the conclusion. That is, there are no differences in the results of the studies, making further analysis redundant as there is nothing to reconcile. However, meta-analysis must always address the possibility of publication bias so that its conclusions can be strengthened.

In the context of the selected dataset, there are a number of reasons why publication bias may be present. The first could be inherent human bias. It is more desirable to researchers and the public alike to find that CBT reduces the odds of recidivism. Thinking logically, society as a whole benefits from a reduction in crime, which is what therapy seeks to achieve. Since rehabilitation is desired, researchers (and more importantly, publishers) may be apprehensive to report contrary results. Another reason is that the publishing of insignificant results can directly affect those providing CBT. That is, the reputation of therapeutic companies may be diminished if studies found that their services were not effective in reducing recidivism. This could dramatically affect their business, and perhaps the industry as a whole, adding to the reluctance to publish insignificant results.

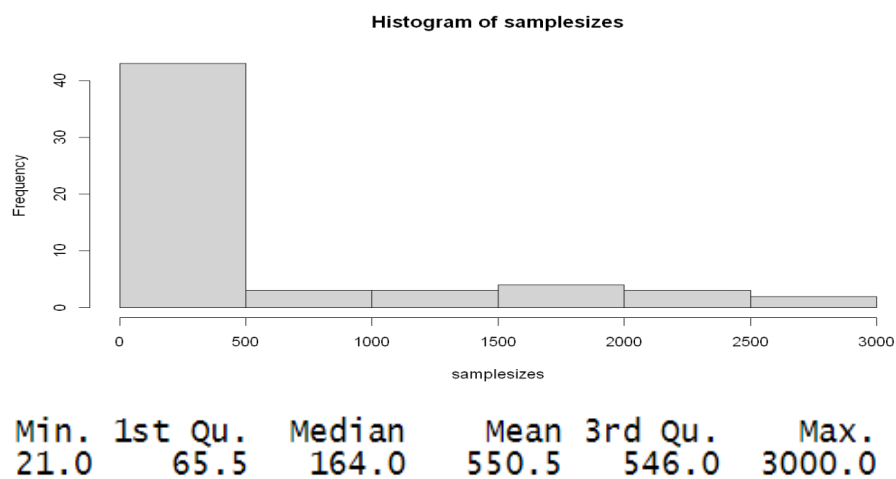
While the purpose of research must always be to provide information that will be useful to society, it is certainly possible for bias to exist. In some cases it may be easy to forget that insignificant results can be just as insightful as significant results. If there is any belief that studies could have been missed systematically within the analysis, an explicit analysis of bias must be conducted. There are several methods within statistical meta-analysis that can address these issues in order to prevent inaccurate conclusions and improve the reliability of its results.

II - Statistical Meta-Analysis

Preliminary Data Exploration

Before performing any meta-analysis techniques, exploration of the data is required to gain a preliminary understanding of the results. The first step would be to establish the need for meta-analysis. As mentioned before, many of the studies found CBT to be effective in reducing recidivism, while some did not. Recall that an effect size greater than zero suggests recidivism is higher in the control group than in the treatment group. Of the 58 studies in the dataset, 6 reported data corresponding to a negative effect size estimate. While this is a relatively small proportion, it is certainly not negligible. The high proportion of positive-effect studies immediately suggests that CBT is an effective means of rehabilitation, but it must also be considered that contrary studies may have been systematically excluded from the overall dataset.

It should be noted that the previous assertions did not consider the statistical significance of any of the results. It is logical to expect the significance to vary for both positive and negative effect sizes, depending upon the sample size of the study in question. That is, larger studies would have more precise effect size estimates, thus having a smaller standard error and in turn a more statistically significant result. For that reason, it will be useful to analyze the total sample sizes of the studies in the dataset, because the standard error of the effect size is directly dependent upon the number of individuals in each group. Below is a histogram of the sample sizes followed by summary statistics.



It is clear that there is a wide range of total sample sizes, with a large mean value of 550.5. This value should be sufficient for reliable results in most cases. However, the vast majority of studies fall into the smallest interval on the histogram. This could be indicative of several insignificant results, as studies with smaller sample sizes are

expected to have wider confidence intervals about their effect size, potentially including zero. Such a confidence interval indicates there is no significant effect of CBT on reducing recidivism.

A study with a smaller sample size is bound to have a larger standard error. They will have less weight in the meta-analysis models then, because the models give greater importance to the studies with more statistically reliable results. For instance, the study with only 21 participants in total is likely to be insignificant, and because of its small sample size is extremely likely to have little to no weight on the meta-analysis models. Conversely, the study with 3000 participants is likely to have a very large impact on the models - its standard error will be dramatically smaller.

Now that the existence of contrasting results and effect of sample size on significance have been established, meta-analysis can be utilized to reconcile any discrepancies.

Fixed-Effect Meta-Analysis

The first method of meta-analysis to be used is known as a fixed-effect model. This model assumes that there is a single common effect size among all of the k studies. That is, any discrepancies between the observed effects are simply due to sampling error within the studies. Mathematically, the model can be written as follows:

$$\hat{\theta}_i = \theta + \varepsilon_i$$

$\hat{\theta}_i$ refers to the observed effect size of the i th study, θ to the true effect size, and ε_i to the sampling error of the i th study. The error term is assumed to follow a normal distribution with mean 0 and variance corresponding to the i th study.

The goal of meta-analysis, as mentioned previously, is to determine a pooled effect size estimate. This estimate is based on a weighted average of the k observed effect sizes, with the weight of each study being dependent upon the variance of its effect size. This pooled effect size is given by the following:

$$M = \frac{\sum_{i=1}^k w_i \hat{\theta}_i}{\sum_{i=1}^k w_i}$$

where $w_i = \frac{1}{V_{\hat{\theta}_i}}$, the inverse of the within-study variance of the i th study

The formulation above gives more weight to studies with lower variances. That is, studies with larger sample sizes will have a greater impact on the overall effect size estimate. This method was implemented using R, its results can be found below.

Fixed-Effects Model (k = 58)

logLik	deviance	AIC	BIC	AICC
-94.6547	213.6898	191.3093	193.3698	191.3808

I ² (total heterogeneity / total variability):	73.33%
H ² (total variability / sampling variability):	3.75

Test for Heterogeneity:
Q(df = 57) = 213.6898, p-val < .0001

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub
0.3382	0.0269	12.5870	<.0001	0.2856	0.3909

The pooled effect size estimate is 0.3382, with a corresponding standard error of 0.0269, giving a test statistic of 12.587. This leads to an extremely small p-value, meaning the effect size is significantly different to 0 at the 99.99% confidence level. Since the value is positive, it can be concluded that cognitive behavioral therapy *is effective* in reducing recidivism among juveniles and adults. It should not

be lost that this result comes from a weighted average with weights depending on within-study variability. That is, the result could be interpreted as proof that the studies with positive effects carried significantly more weight than those with negative effects. Or, the number of positive effect studies may have simply exceeded the number of negative effect studies, bringing the overall average further from zero in the positive direction. This ties back into the issue of publication bias in that studies with negative effects may have been missed in the dataset. If these studies were included, it is possible that the results would change, albeit unlikely given the high confidence of parameter significance.

For a clearer interpretation of the results, the pooled effect size can be exponentiated to obtain 1.402, meaning that non-treated individuals are 1.402 times more likely to recommit than those treated with CBT.

The previous output includes several “heterogeneity” estimates in addition to the fixed-effect model results, referring to the variability between studies. The fundamental assumption behind fixed-effect models that there is only one true effect size can be problematic for a number of reasons. Studies conducted at different times or at different locations, or with slightly different designs or participants may not be directly comparable. These factors can be sources of the aforementioned “heterogeneity”, or the between-study variability. In cases where the fixed-effect assumption is not appropriate, each study has its own effect size, meaning it is now a random variable. This brings about the existence of random effects meta-analysis models, which take the between-study variability into account.

Random Effects Meta-Analysis

If there is any belief that there is no common effect size among all studies, - or, in other words, that there is a significant source of variation between studies - then random effects meta-analysis should be utilized. The results from the fixed-effect model on the previous page include the following statistics: I^2 , H^2 , and Q . All three are indicators of heterogeneity, and can establish the need for a random effect model. Explanations of these parameters are below.

- I^2 : the proportion of the total variability that is between-study variability. In other words, it is the total heterogeneity divided by the within-study variability plus the between-study variability. The value in the output is 73.33%, meaning nearly three-quarters of the total variability is due to heterogeneity.
- H^2 : the ratio of total variability to sampling variability. The value is 3.75, meaning there is 3.75 times as much total variability as sampling variability. A value closer to 1 would indicate that the sampling variability is very close to the total variability.
- Q : the weighted sum of squares between the pooled effect size estimate and the effect size of each study. If this value is large, it means that the individual effect sizes are greatly dispersed about the pooled estimate. The corresponding value for this dataset is 213.68, with an associated p-value smaller than 0.0001, meaning it is significantly different to 0, so there is statistically significant heterogeneity in the dataset.

Each of the above parameters suggests there is a large amount of between-study variability, justifying the necessity of a random effects model. These models seek to estimate the expected value or the mean of the effect sizes. Its mathematical representation follows:

$$\hat{\theta}_i = \mu + \xi_i + \varepsilon_i$$

$\hat{\theta}_i$ once again refers to the observed effect size of the i th study, μ to the overall mean of the true effects θ_i , ξ_i to the effect of the i th study to the overall mean, and ε_i to the difference between the observed effect and its respective true effect.

Random effects models have two sources of variability: within-study and between-study. The former is readily calculable using the standard error formula on page 4, whereas there are various means of calculating the latter. The following results used the DerSimonian-Laird method of calculating heterogeneity.

The pooled effect size estimate is calculated similarly to the random effects model, the only difference being the inclusion of the τ^2 parameter for the between-study variability.

$$M^* = \frac{\sum_{i=1}^k w_i^* \hat{\theta}_i}{\sum_{i=1}^k w_i^*}$$

where $w_i^* = \frac{1}{V_{\hat{\theta}_i}^*}$, the inverse of the total variability of the i th study. More concretely,

$V_{\hat{\theta}_i}^* = Var(\hat{\theta}_i) + \tau^2$, the within-study variability plus between-study variability

The inclusion of an extra parameter for variance has several implications on the results. Firstly, it is expected for all confidence intervals to be wider. That is, the confidence interval around the effect size for each individual study and for the weighted average will be wider because there are now two sources of variability. Additionally, each study will carry a more equal weight, as each respective weight is being scaled by the same value. The larger the estimate for τ^2 , the more equal the weights of each study will be. This parameter was calculated and the model was implemented using R, the results are as follows.

```
Random-Effects Model (k = 58; tau^2 estimator: DL)

logLik deviance AIC BIC AICC
-42.5827 109.5459 89.1654 93.2863 89.3836

tau^2 (estimated amount of total heterogeneity): 0.1196 (SE = 0.0445)
tau (square root of estimated tau^2 value): 0.3458
I^2 (total heterogeneity / total variability): 73.33%
H^2 (total variability / sampling variability): 3.75

Test for Heterogeneity:
Q(df = 57) = 213.6898, p-val < .0001

Model Results:

estimate se zval pval ci.lb ci.ub
0.4271 0.0631 6.7633 <.0001 0.3033 0.5508 ***

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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
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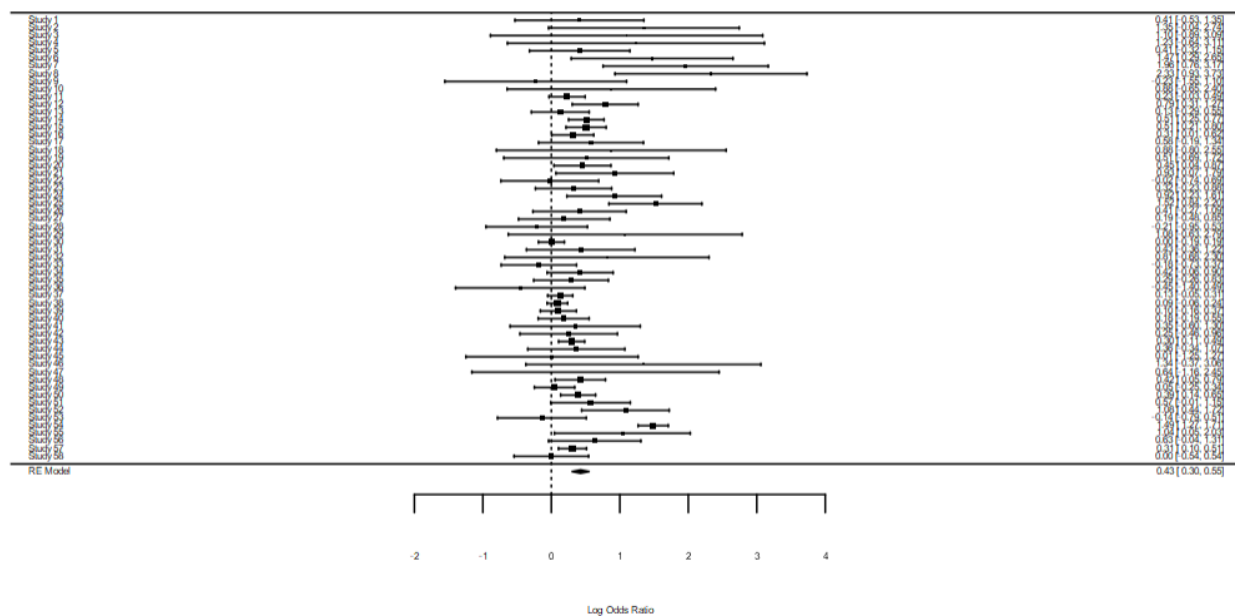
The estimate for the mean of effect sizes was calculated to 0.4271 with standard error 0.0631. This gives a large test statistic, and in turn a very small p-value, once again suggesting that there is a higher proportion of non-recidivists in the CBT group than in the control group. That is, the results support the conclusion that cognitive behavioral therapy is effective in

reducing the tendency of criminals to recommit crimes. τ^2 was estimated to be 0.1196 using the DerSimonian-Laird Method. This value may be difficult to interpret as it simply refers to the amount of total heterogeneity, but using its standard error of 0.0445, a test statistic of 2.6876 can be acquired. This has a p-value of 0.003597947, meaning it is significantly different to 0 and should thus be included in a meta-analysis model. The necessity for a random effects model is additionally supported by the estimates for I^2 , H^2 , and Q as discussed before.

It should also be noted that although the p-value regarding the significance of the overall effect is the same as the previous model, the standard error of the estimate is larger. Thus, the confidence interval is wider and the test statistic is smaller. This is a direct result of the extra source of variability given by τ^2 . The overall effect size estimate and its lower confidence

interval limit are greater than their respective fixed-effect estimates, meaning that an even greater difference between the control and treatment group exists. That is, the random effects model suggests even more so that CBT reduces recidivism because its pooled estimates are larger. Once again the result can be readily interpreted by exponentiating the overall effect estimate. This gives a value of 1.5328, meaning participants not treated with CBT are 1.5328 times more likely to recommit than participants treated with CBT. This value is larger than the exponentiated value in the fixed effect model, hence random effects detected a greater difference between the groups.

A useful method of visualizing the results of a meta-analysis is a forest plot. The following plot shows the effect size estimates of each study along with horizontal lines to represent the confidence interval around them. The size of the squares represent the “influence” or the weight of the study on the overall estimate. Thus, larger squares were more impactful on the pooled effect than smaller squares. The diamond at the bottom of the plot represents the random effects model - its effect size estimate and confidence interval.



Many of the studies have confidence intervals that cross over 0, suggesting that their results are not significantly different to 0. However, there are no studies with confidence intervals that are entirely below 0, which contributes to the overall effect being greater than and significantly different to 0. Also contributing to this are studies with large weights and significant results, namely studies 13, 14, 54, and 57.

Overall, the results from both the fixed-effect and random effects models were very similar. Both models led to the conclusion that cognitive behavioral therapy reduces the odds of recidivism. This can serve as an answer to the research question. Despite there being some studies to the contrary, and the fact that many studies had insignificant results, there is sufficient evidence among all of the given studies to suggest that CBT is effective in reducing recidivism.

Also worth noting is that the convention is to begin with random effects models and reduce to fixed-effect models if appropriate, but this order was reversed for demonstration purposes. Additionally, it is possible for the results of the random effects model to vary depending upon the method used to estimate τ^2 , but these were excluded because the intention is to establish the background of meta-analysis with an emphasis on the effects of publication bias.

Meta-Regression

As discussed in the introduction, the dataset includes several additional variables outside of those involved in the calculation of the odds ratio. More specifically, it includes variables corresponding to the demographics of the participants, the length and regularity of the therapy sessions, and specific techniques utilized within CBT. It is reasonable to assume that at least one of these variables has an effect on the outcome of interest. For instance, one might expect a study where patients participated in more therapy sessions to have a larger effect size because therapy can require time before any significant change is seen. Conversely, a shorter study may not have given their patients sufficient time to rehabilitate, so the resulting odds of recidivism may be close to equal for both the treated and control groups. The potential of confounding variables affecting the results of the analyses can be assessed through meta-regression.

Meta-regression is very similar to regression in its simplest terms. The procedure treats the effect size as the response variables, and the potential confounds as the explanatory variables. The purpose is to ascertain whether these confounds, or moderators, are significant in explaining the variance between the observed effect sizes. As with the previous analyses, this technique was implemented using R. The potential covariates tested as moderators for the odds ratio were: average participant age, the proportion of male participants, the total hours of therapy conducted, and a binary variable indicating whether the anger control component of CBT was utilized. While there are certainly more potential covariates in the dataset, these seem as though they are intuitively related to the outcome. Additionally, the objective is simply to demonstrate a method of explaining the variability between studies. It should also be noted that some of these variables were missing values, if this was the case then the program removed the study from the model fitting, decreasing the sample size slightly.

All Moderators

Warning: Studies with NAs omitted from model fitting.

Mixed-Effects Model (k = 43; tau² estimator: REML)

logLik	deviance	AIC	BIC	AICc
-32.6917	65.3833	77.3833	87.2088	80.0930

tau² (estimated amount of residual heterogeneity): 0.1508 (SE = 0.0628)

tau (square root of estimated tau² value): 0.3883

I² (residual heterogeneity / unaccounted variability): 65.24%

H² (unaccounted variability / sampling variability): 2.88

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 38) = 128.6929, p-val < .0001

Test of Moderators (coefficients 2:5):

QM(df = 4) = 1.6094, p-val = 0.8071

	estimate <chr>	se <chr>	zval <chr>	pval <chr>	ci.lb <chr>	ci.ub <chr>
intrcpt	0.6747	0.4600	1.4666	0.1425	-0.2270	1.5764
covarsage	-0.0088	0.0135	-0.6535	0.5135	-0.0353	0.0176
covarsmale	-0.0008	0.0035	-0.2248	0.8221	-0.0077	0.0061
covartotalhrs	0.0001	0.0004	0.3627	0.7168	-0.0006	0.0008
covarsanger	0.1697	0.1686	1.0065	0.3142	-0.1607	0.5000

The model with all four moderators was not able to explain any of the heterogeneity between the studies. Each parameter estimate was extremely close to zero, and thus had a small test statistic and large corresponding p-value. This suggests that none of them are significantly different to 0, and have no significant effect on the differences observed in the effect sizes. This could be a result of multicollinearity, where at least one of the explanatory variables is redundant of the others. This leads to higher standard errors for the parameter estimates, thereby decreasing their statistical significance. Testing the moderator parameters individually reveals similar results for age, male proportion, and hours of therapy, with p-values of 0.6290, 0.6563, and 0.6539, respectively. All of these are very insignificant. However, testing the anger control component individually gave a parameter estimate p-value of 0.0331, significant at $\alpha = 0.05$. This means that the inclusion of the anger control component in CBT significantly affects the proportion of recidivists in the treatment group. The results of this model are as follows.

CBT Anger Control Component Only

Mixed-Effects Model (k = 58; tau² estimator: REML)

logLik	deviance	AIC	BIC	AICc
-40.0462	80.0924	86.0924	92.1685	86.5539

tau ² (estimated amount of residual heterogeneity):	0.0895 (SE = 0.0319)
tau (square root of estimated tau ² value):	0.2991
I ² (residual heterogeneity / unaccounted variability):	66.55%
H ² (unaccounted variability / sampling variability):	2.99
R ² (amount of heterogeneity accounted for):	14.49%

Test for Residual Heterogeneity:
QE(df = 56) = 175.9401, p-val < .0001

Test of Moderators (coefficient 2):
QM(df = 1) = 4.5431, p-val = 0.0331

Model Results:

	estimate <chr>	se <chr>	zval <chr>	pval <chr>	ci.lb <chr>	ci.ub <chr>
intrcpt	0.3322	0.0701	4.7366	<.0001	0.1947	0.4696
covars[, 4]	0.2618	0.1228	2.1315	0.0331	0.0211	0.5025

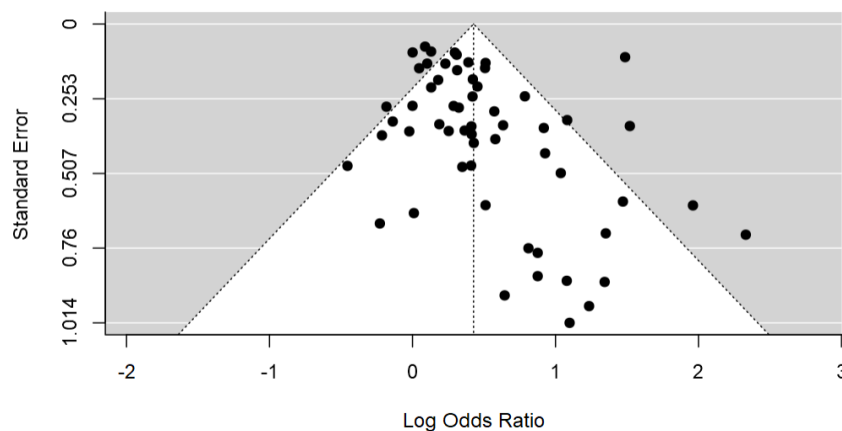
The parameter estimate is 0.2618, meaning the log-transformed odds ratio increases when the variable is equal to 1. In other words, the proportion of non-recidivism increases for individuals who received treatment involving the anger control component. The estimate also has a wide confidence interval, so the effect could be minimal, but it is nonetheless statistically significant. The parameter estimate for the intercept of the model is of course significant, which was to be expected. This is because the results of both meta-analysis models found the pooled effect to be significantly different to 0, then it follows that the intercept of a regression model will not be at the origin. The R^2 value is 14.49, meaning the moderator explained 14.49% of the heterogeneity between studies. While this is not a particularly high amount, the results above could inform researchers on CBT that the anger control component is significant in reducing the chances of recidivism.

III - Publication Bias

All of the previous results, while statistically significant, can be rapidly diminished in reliability if not invalidated entirely. This would be a result of publication bias. As mentioned before, publication bias is the tendency of researchers and publishers to avoid publishing non-significant or undesirable results. While the forest plot in the random effects method revealed that many of the effect sizes were insignificant, there is still the possibility that studies with negative effect sizes were omitted. This especially must be considered because of the high proportion of positive effects reported. Even if small effect sizes were not systematically missed, the following procedures are necessary in statistical meta-analysis to prove the validity of results. This will also provide complete context to the problem, and potentially establish the need for further research on the effect of CBT on recidivism.

Funnel Plots

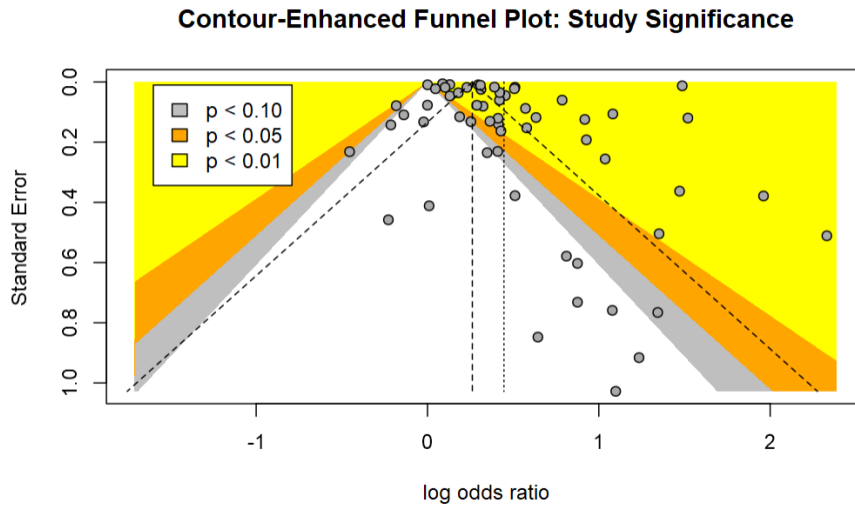
The first method to examine publication bias is a funnel plot, which plots every effect size in the analysis on the x -axis along with its corresponding standard error on the y -axis. Since the effect size is a log-transformed odds ratio, the vertical axis will be inverted. Recall that the standard error of the study depends directly upon the sample size of the study - larger studies will have a smaller standard error. Ideally, the plot will be symmetrical about the overall effect size estimate. This means that for every large study with a large effect size, there will be a corresponding large study with a large effect size. Or, for every small study with a large effect size, there will be a corresponding small study with a large effect size. Essentially, every study would be approximately “mirrored” by another study in the dataset. The studies are expected to follow a funnel shape because detecting a larger difference requires less precision. Thus, studies with higher precision are expected to be closer to the overall estimate. Below is the funnel plot for the cognitive behavioral therapy dataset. The dotted line in the center represents the weighted effect size estimate, and the points represent each individual study.



Clear asymmetry can immediately be seen in this instance, as the studies to the right of the center line are far more dispersed than those to the left of the line. In other words, studies below the weighted effect size estimate of 0.4271 are grouped closely, meaning their standard errors are all relatively small. The studies above this

estimate have a much wider range of standard errors - some are very small and others are very large. As mentioned in the exploratory data analysis section, there is an extremely wide range of study sizes in the dataset, explaining the dispersion of the points in the plot. Hence, many of the studies do not have a mirrored counterpart. This is primarily illustrated by the studies with very large effect sizes and relatively small standard errors. As mentioned, it requires more precision to

detect a smaller difference, so these studies contradict this intuition and fall outside the expected funnel shape. For clarity, these studies detected a very large difference despite having a small standard error. This asymmetry reveals the possibility of publication bias.



The plot to the left is a further visualization of publication bias, with colored contours included to show the statistical significance of each study. The programming in R created the funnels according to the expected precision required to detect the given effect size. As mentioned, detecting a larger difference requires a larger

standard error. Both of which are augmented further at higher levels of significance. It can be seen that there is a very large quantity of studies to the right of the weighted estimate with statistically significant results at $\alpha = 0.01$, and virtually none to the left of the weighted estimate. While this further demonstrates asymmetry, it shows that most of the large studies have very significant positive effect sizes. Moreover, many of these studies are significant at confidence levels exceeding 99.99999%, meaning there is an extremely high probability the effect size is in fact greater than 0. The most important observation is that there are no equivalent highly significant large studies to the left of the weighted effect size estimate. This plot also reflects the results seen in the random effects forest plot - most of the studies had relatively small standard errors but effect sizes close to zero and thus insignificant results. However, a number of the studies had *very* significant results *far* from zero. The lack of significant studies with negative effect sizes could indicate that CBT is strongly associated with a reduction in recidivism, despite the asymmetry suggesting bias. This cannot be concluded without conducting explicit tests to address it.

Egger's Regression Test

The first explicit test is Egger's Regression. This is a formal test for asymmetry in the funnel plot, and thus a mathematical representation of publication bias. It consists of a simple linear regression model of the following form:

$$\frac{\hat{\theta}_k}{SE(\hat{\theta}_k)} = \beta_0 + \beta_1 \left(\frac{1}{SE(\hat{\theta}_k)} \right)$$

The standardized effect size of the k th study serves as the response variable, while its corresponding precision serves as the explanatory variable. Since the effect size in question is the log-transformed odds ratio, the model can be re-written as:

$$\frac{\hat{\theta}_k}{\sqrt{1/A+1/B+1/C+1/D}} = \beta_0 + \beta_1 \left(\frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D} \right)^{-1/2}$$

A symmetrical funnel plot would lead to a parameter estimate near zero for β_0 . In other words, the regression line should pass through the origin. This is because the intercept parameter is a reflection of the effect sizes when all other explanatory variables are zero. Thus, the intercept shows the parameter estimate when the precision is zero, i.e. the standard error is infinitely large. This should be around zero when there is no publication bias, because studies with larger standard errors have wider confidence intervals, decreasing the chances of being significantly different to zero. When there *is* publication bias, smaller studies with large effect sizes may be overrepresented, increasing the chances of being significantly different to zero. In this model, the parameter estimate for β_1 is inconsequential.

The model will once again be entirely dependent upon the sample sizes. Since there is a very wide range of sample sizes, intuition would suggest that the test will reveal asymmetry. The results were calculated and are shown below.

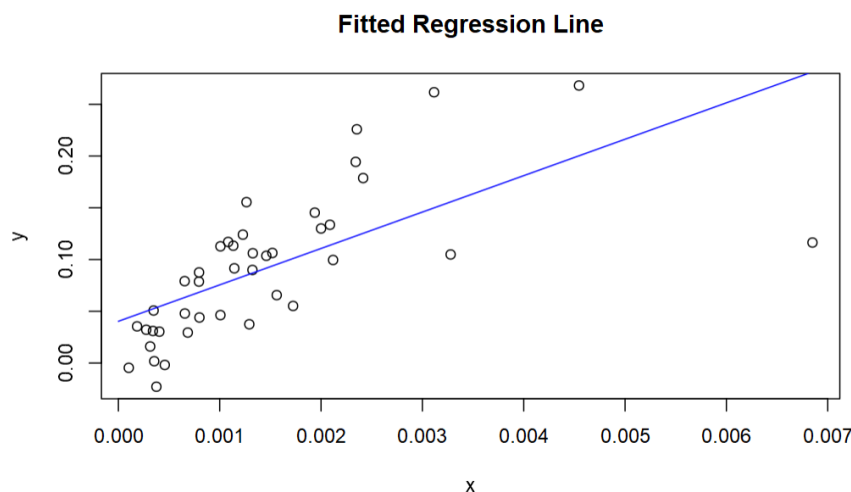
The output from R shows that the estimate for β_0 is 0.04036, which is very small, but with a small standard error it corresponds to a large test statistic and ultimately a small p-value. The parameter is significantly different to zero at the 99% confidence level, meaning there is significant asymmetry in the funnel plot, suggesting publication bias.

```
Call:
lm(formula = y ~ x, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.164902 -0.029708  0.003049  0.033129  0.111718

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.04036    0.01223   3.301  0.00207 **
x           35.18325    6.39764   5.499  2.57e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05191 on 39 degrees of freedom
Multiple R-squared:  0.4368,    Adjusted R-squared:  0.4223 
F-statistic: 30.24 on 1 and 39 DF, p-value: 2.565e-06
```



The fitted regression line also demonstrates these findings. Although the data exists on a very small scale, the line clearly does not intersect the y-axis close to zero relative to the slope. Generally, these results indicate there may be some instances of the small-study effect in the dataset. This occurs when small studies have larger effect sizes than large studies. These studies are expected to have a higher

standard error, and would thus carry less weight in the meta-analysis procedures, but they can still reduce the validity of results if they outnumber the large studies. This was clearly demonstrated by the histogram in the exploratory section, where the vast majority of the studies

were relatively small compared to the mean size. That said, the primary cause of this result is likely the outlying data point on the far right, which effectively “pulls” the regression line down. This study had the effect of increasing the intercept parameter estimate, making it significantly different to zero.

Trim-and-Fill Method

Now that the potential existence of publication bias in the dataset has been established, methods of addressing it can be implemented. The primary method to be utilized is the trim and fill method, which consists of two parts as its name suggests. The first part is trimming, in which outlying studies are identified and removed from the analysis. Next, the overall effect size is recalculated, typically with a fixed effect model. The new pooled effect estimate is used as a “mirror” in the center of the funnel plot. This brings about the next part, filling. Using the aforementioned mirror, hypothetical studies are added to the dataset mirroring their trimmed counterpart. For instance, if the dataset had an outlying study with an effect size of 0.4 and standard error of 1, its filled counterpart would have an effect size of -0.2 and the same standard error of 1, assuming the pooled effect estimate is 0.1. The final step is to recalculate the overall effect size with all of the trimmed studies and filled studies. It should be mentioned that the results of the trim and fill method can be unreliable with high heterogeneity, because the studies do not share a single true effect size. This is in violation of the fixed effect model assumption, and applies to this example because of the high between-variability of the studies. Once again, this is primarily a result of the large range and variance of sample sizes utilized in the studies. However, the technique can still be implemented for demonstration purposes.

The following images show the trim-and-fill method results. The summary of the results indicates that 16 studies were added to the left side of the funnel plot. Since they were added to

```
Estimated number of missing studies on the left side: 16 (SE = 4.9484)
Random-Effects Model (k = 74; tau^2 estimator: DL)

logLik deviance AIC BIC AICC
-78.0518 169.6780 160.1036 164.7117 160.2726

tau^2 (estimated amount of total heterogeneity): 0.2320 (SE = 0.0767)
tau (square root of estimated tau^2 value): 0.4817
I^2 (total heterogeneity / total variability): 82.11%
H^2 (total variability / sampling variability): 5.59

Test for Heterogeneity:
Q(df = 73) = 407.9530, p-val < .0001

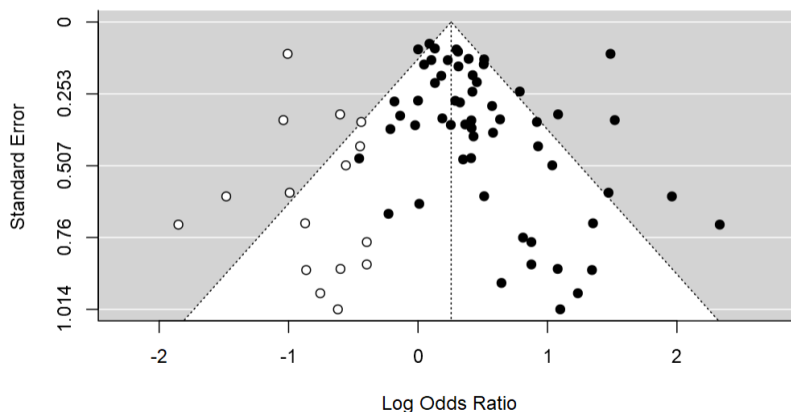
Model Results:

estimate se zval pval ci.lb ci.ub
0.2557 0.0730 3.5030 0.0005 0.1126 0.3988 ***

---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

that CBT reduces recidivism, because the inclusion of many negative log-odds ratios had little to no effect on the conclusion. Also worth noting is that the heterogeneity increased, which is to be expected when adding studies to the analysis. An image

the *left* side, this means that a large number of insignificant studies were imputed. This was to be expected due to the un-mirrored studies to the right of the original funnel plot. Interestingly, the results of the method show that the overall effect size estimate is still significantly different to zero at the 99.99% confidence level, albeit with a wider confidence interval. This indicates that the original dataset has overwhelming evidence



of the final funnel plot is shown above for visualization purposes. It is clearly more symmetrical, and would likely lead to a less significant intercept estimate in Egger's Regression test. The plot shows the addition of several negative effect sizes, each mirroring positive effects which were determined to be outlying results in the original data.

File-Drawer Problem

The final area of consideration is the file-drawer problem. Common in psychological or behavioral meta-analyses, it is based on the fact that studies with large effect sizes are more likely to be published than studies with small effect sizes. The method associated with it involves combining the p-values from each of the k studies to determine whether k_0 unpublished non-significant studies exist. Inverse normal methods are used to combine the p-values, with

$$Z = \frac{1}{\sqrt{k}} \sum_{i=1}^k \Phi^{-1}(P_i), \text{ where } P_i \text{ is the p-value from the } i\text{th study.}$$

The null hypothesis that the overall effect size is significant is rejected if $|Z| > z_{1-\alpha}$. In context, if the combined p-value is larger than the cut-off point, there is a significant effect of cognitive behavioral therapy on reducing recidivism.

The above equation can be used to calculate k_0 :

$$k_0 > -k + \left(\frac{\sum_{i=1}^k z_i}{z_{1-\alpha}} \right)^2$$

the minimum number of non-significant studies required to change the overall conclusion from significant to non-significant. This is easily calculated using R with the Fail-safe N calculation, the results of which can be found below.

Clearly, there would have to be an extremely large number of insignificant studies for the overall conclusion to change. While this suggests that publication bias may not be a major

Fail-safe N Calculation Using the Rosenthal Approach

Observed Significance Level: <.0001

Target Significance Level: 0.05

Fail-safe N: 3249

problem with the dataset, it is always important to address the possibility of bias. These results directly agree with the results of the trim-and-fill method, which gave a very

significant positive effect size despite 16 insignificant studies being added to the dataset. Regardless of this result, it was best to maintain conservatism by adjusting the dataset via the trim and fill method. A large number of studies reported a significant positive effect size, which may have been due to publication bias but more likely would have been due to the heavily supported conclusion that CBT is effective in reducing recidivism. In all, this result is possibly the strongest evidence that CBT should be used to effectively rehabilitate offenders.

IV - Discussion and Conclusion

The procedures and methods in this report have outlined the techniques of meta-analysis along with checks on the reliability of results. Specifically, it was found that cognitive behavioral therapy is significant in reducing the odds of recidivism in juveniles and adults. This result was shown through the implementation of a fixed-effect and a random effects model, both of which generated a significant and positive overall effect size estimate. Due to the significant heterogeneity found between the studies, the random effects model would be selected as a more appropriate representation of the dataset. That is, it cannot be reliably assumed that there is a single common effect size among the studies, likely a direct result of the wide range of sample sizes utilized. Exponentiating the final pooled effect size estimate gives a value of 1.5328, meaning individuals not treated with CBT have 1.5328 times the odds of re-committing crimes than individuals who received therapy. Hence, the use of CBT is effective in rehabilitation. Additionally, the meta-regression analysis found that the anger management component of CBT is particularly effective in reducing recidivism, a discovery that may influence researchers to place more emphasis on including the component in therapy sessions.

This conclusion would not be valid if publication bias were not addressed. Although the funnel plot showed considerable asymmetry and Egger's Regression test calculated a significant parameter estimate for the intercept, publication bias may not be a significant factor in the selected dataset. This is because the trim-and-fill method had almost no effect on the conclusion despite 16 studies with negative effect sizes being imputed. Additionally, the Fail-safe N calculation to address the file-drawer problem revealed that more than three thousand insignificant studies would have to be included to change the overall conclusion to insignificant. This means that the relationship between CBT and recidivism is thoroughly and accurately represented by the studies in the dataset. Though both of these results demonstrate that publication bias is not a problem in this case, the conclusion drastically loses validity if methods were not implemented to *prove* the lack of bias. For the question at hand, these results greatly support the conclusion that cognitive behavioral therapy should be used as a means of rehabilitating offenders.

Without addressing bias, it may have been inaccurately concluded that the results showed a positive relationship between CBT and criminal rehabilitation. However, the exploration of bias confirmed the results shown by many of the studies. In particular, the trim-and-fill method and Fail-safe-N calculation proved that there is overwhelming evidence in favor of cognitive behavioral therapy reducing recidivism. Generally, it is imperative that publication bias is addressed in statistical meta-analysis and in statistics generally. Otherwise, inaccurate conclusions can very easily be made and accepted as truth. At best, a failure to address publication bias sets back the understanding of the subject matter at hand, either by leaving out important information or failing to validate findings. At worst, such a failure can negatively affect the lives of individuals or society as a whole. The latter is something to be avoided at all costs, thus the addressing of publication bias is imperative to *any* analysis in *any* field.

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Appendix

Package / Data Loading and Exploration

```
library(meta)
library(metadat)
data("dat.landemberger2005")
cbt = dat.landemberger2005
samplesizes = cbt[,8]+cbt[,9]+cbt[,10]+cbt[,11]
hist(samplesizes)
summary(samplesizes)
var(samplesizes)
```

Meta-Analysis Procedures

```
library(metafor)
eff.size.dat = escalc(measure="OR", ai=n.cbt.non, bi=n.cbt.rec, ci=n.ctrl.non,
                      di=n.ctrl.rec, data=cbt)
fixed_analysis = rma(yi,vi,method='EE',data=eff.size.dat)
summary(fixed_analysis)
random_analysis = rma(yi,vi,method='DL',data=eff.size.dat)
summary(random_analysis)
pnorm(2.68764,lower.tail=F)
forest(random_analysis)
```

Meta-Regression

```
age = cbt[,14]; male = cbt[,15]; totalhrs = cbt[,20]; anger = cbt[,25]
for (i in 1:length(anger)) { # reformatting anger management variable
  if (anger[i]=='yes') {anger[i] = 1}
  if (anger[i]=='no') {anger[i] = 0}}
covars = cbind(age,male,numsess,totalhrs,anger)
covars = apply(covars,2,as.numeric)
cat('All Moderators \n')
model1 = rma(yi,vi,mods=~covars,data=eff.size.dat)
summary(model1)
cat('Age Only \n')
model2 = rma(yi,vi,mods=~covars[,1],data=eff.size.dat)
summary(model2)
cat('Male Proportion Only \n')
model3 = rma(yi,vi,mods=~covars[,2],data=eff.size.dat)
summary(model3)
cat('Total Hours of Therapy Only \n')
model4 = rma(yi,vi,mods=~covars[,3],data=eff.size.dat)
summary(model4)
cat('CBT Anger Control Component Only \n')
model5 = rma(yi,vi,mods=~covars[,4],data=eff.size.dat)
summary(model5)
```

```

# Publication Bias
funnel(random_analysis)
colors = c("gray75", "orange", "yellow")
analysis = metagen(yi,vi,data=eff.size.dat,random=T,method.tau="DL")
funnel.meta(analysis, contour = c(0.90, 0.95, 0.99),col.contour = colors, xlab='log odds ratio')
legend(x = -1.6, y = 0.01, legend = c("p < 0.10", "p < 0.05", "p < 0.01"), fill = colors)
title("Contour-Enhanced Funnel Plot")

y = yi/vi # set up variables for quantifiable test
x = 1/vi
eggertest = lm(y~x,data=data) # egger's test
summary(eggertest) # B0 significant
plot(x,y)
xvec = seq(0,0.008,by=0.0001)
betas = eggertest$coefficients
lines(xvec,betas[1]+betas[2]*xvec,col='blue') # superimpose fitted line
title('Fitted Regression Line')

tf = trimfill(random_analysis)
summary(tf)
funnel(tf) # trimmed and filled plot

fsn(yi,vi,data=eff.size.dat)

```